

Revised Generation Interconnection System Impact Study Report

For Queue Project Q12-0137

Gates 100 kV
June 26, 2014

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q12-0137. The Interconnection Customer proposes a 105.3 MW wind facility interconnecting at the Gates 100 kV substation in the Indiana area. The request was received on June 13, 2012. The study evaluated system intact and single contingency conditions using the regional planning case, including 91 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 17 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$33,681,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

The Pueblo Xtra 500/230 kV XFMR is overloaded to 129.5% of its 766 MVA emergency rating for the loss of the Boca Raton-Pueblo Xtra 500 kV. The project contributes 19.6% DFAX (20.7 MW) to this overload.

For the loss of Hilltop-Piney Crk Project 500 kV line, the Piney Crk Project 500-345 kV Transformer loads to 124.6% of its 776 MVA rating. Project DFAX: 9.9%.

The Adams-Onancock Diesel Generation Facility 230 kV is overloaded to 120.4% of its 670 MVA emergency rating for the loss of the Trego-Beaver 500 kV. The project contributes 17.1% DFAX (18.0 MW) to this overload.

Larson Street-Norwich 230 kV: 114.8% loading (559 MVA) following the outage of the Barrett-Norwich 500 kV; DFAX = 9.8%.

For the loss of Beaver-Eufaula 500 kV, the Beaver-Onancock Diesel Generation Facility 230 kV loads to 113.5% of its 605 MVA rating. Project DFAX: 11.9%.

The Ft Humbug 500/345 kV Transformer Bank is overloaded to 111.8% of its 628 MVA emergency rating for the loss of the Orchard Rd-Fort Humbug 500 kV L. The project contributes 5.2% DFAX (5.5 MW) to this overload.

The Little Chute-Talenenergy Martins Crk Llc Harrisburg Substation 345 kV L is overloaded to 107.0% of its 1,012 MVA emergency rating for the loss of the Coronet-Fort Humbug 500 kV L. The project contributes 7.6% DFAX (8.0 MW) to this overload.

The Hillcrest Substation-Segs 5 Substation 100 kV is overloaded to 105.6% of its 129 MVA emergency rating for the loss of the Richland-Dekalb 500 kV. The project contributes 7.7% DFAX (8.1 MW) to this overload.

The Post Rock-Newport Tie 230 kV is overloaded to 105.3% of its 543 MVA emergency rating for the loss of the Trego-Beaver 500 kV line. The project contributes 14.4% DFAX (15.2 MW) to this overload.

Myrtle Point Substation-Post Rock 230 kV line: 104.6% loading (621 MVA) following the outage of the Trego-Beaver Substation 500 kV; DFAX = 13.9%.

For the loss of Hilltop Substation 500 kV/345 kV Transformer, the Talenenergy Martins Crk Llc Williamsport 500-345 kV XFMR loads to 104.2% of its 821 MVA rating. Project DFAX: 7.0%.

The Beaver 230 kV/1 kV Transformer is overloaded to 102.6% of its 1,021 MVA emergency rating for the loss of the Eufaula-Beaver 500 kV. The project contributes 15.0% DFAX (15.8 MW) to this overload.

For the loss of Beaver-Eufala Substation 500 kV, the Joso-Onancock Diesel Generation Facility 230 kV loads to 102.4% of its 537 MVA rating. Project DFAX: 15.5%.

The Laureles-Ward Gulf 100 kV is overloaded to 101.3% of its 175 MVA emergency rating for the loss of the Dekalb-Jones Chapel 500 kV Line. The project contributes 8.6% DFAX (9.1 MW) to this overload.

Ligonier Substation-Montibello 100 kV: 101.2% loading (136 MVA) following the outage of the Dekalb-Jones Chapel 500 kV; DFAX = 8.6%.

Cypress Crk-Big Oaks 100 kV L: 100.4% loading (136 MVA) following the outage of the Dekalb-Jones Chapel 500 kV; DFAX = 8.6%.

For the loss of Blackrock-Shawville Sta 345 kV L, the Carbon Center 500/345 kV Transformer Bank loads to 100.0% of its 523 MVA rating. Project DFAX: 7.4%.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Pueblo Xtra 500-230 kV Autotransformer	Replace transformer with higher-rated unit	\$24,970,000	2% (\$637,000)
Piney Crk Project 500 kV/345 kV Autotransformer	Replace transformer with higher-rated unit	\$21,939,000	3% (\$660,000)
Adams-Onancock Diesel Generation Facility 230 kV	Rebuild line with higher-capacity conductor	\$76,909,000	3% (\$2,279,000)
Larson St-Norwich 230 kV L	Rebuild line with higher-capacity conductor	\$20,861,000	3% (\$593,000)
Beaver-Onancock Diesel Generation Facility 230 kV	Rebuild line with higher-capacity conductor	\$20,531,000	3% (\$604,000)
Ft Humbug 500 kV/345 kV TX	Replace transformer with higher-rated unit	\$25,435,000	100% (\$25,435,000)
Talenenergy Martins Creek Llc Harrisburg-Little Chute 345 kV	Reconductor line	\$7,221,000	3% (\$220,000)
Hillcrest Substation-Segs 5 Substation 100 kV	Reconductor line	\$782,000	3% (\$26,000)
Post Rock-Newport Tie 230 kV	Reconductor line	\$12,995,000	3% (\$380,000)
Post Rock-Myrtle Pt 230 kV line	Reconductor line	\$10,181,000	3% (\$297,000)
Talenenergy Martins Crk Llc Williamsport 500-345 kV XFMR	Replace transformer with higher-rated unit	\$27,282,000	3% (\$833,000)
Beaver 230 kV/1 kV Transformer Bank	Replace transformer with higher-rated unit	\$7,408,000	3% (\$218,000)
Joso-Onancock Diesel Generation Facility 230 kV	Reconductor line	\$9,835,000	3% (\$291,000)
Ward Gulf Substation-Laureles 100 kV	Reconductor line	\$1,188,000	5% (\$62,000)
Montibello-Ligonier 100 kV	Reconductor line	\$1,948,000	5% (\$102,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Big Oaks-Cypress Crk 100 kV Line	Reconductor line	\$2,157,000	5% (\$113,000)
Carbon Center 500/345 kV Transformer Bank	Replace transformer with higher-rated unit	\$30,768,000	3% (\$931,000)
	Total allocated to project		\$33,681,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.